

Model- and Data-Based Approaches to Wind Compensation for Airship Drones

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Abstract

In here, you should briefly describe the thesis content in the primary language.

Kurzfassung

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Acknowledgements

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Nomenclature

Abbreviations

COG	Center of Gravity
COV	Center of Volume
CS	Coordinate System
DFF	Disturbance Feedforward
EKF	Extended Kalman Filter
KF	Kalman Filter
NDI	Nonlinear Dynamic Inversion
NED	North-East-Down convention
UKF	Unscented Kalman Filter

Latin Letters

${}^{bm}I$ Inertia Matrix with respect to the COG

Greek Letters

ω Angular velocity (rad s^{-1})

Indices

$(\cdot)_B$	Quantity describing the body
${}_B(\cdot)$	Quantity in body-fixed coordinate system
${}_E(\cdot)$	Quantity in Earth coordinate system

1. Introduction

This chapter serves as an overview of existing methods and ...

1.1. Problem setting

1.2. Aim of this work

The Aim of this work is to implement an algorithm for wind estimation and compensation for an unmanned airship. The algorithm should be able to reliably estimate wind speeds in conditions relating to real flight scenarios. Those include: Dynamic wind conditions with Wind gusts of max 30 km/h. Trajectories like forward and sideward flight. Rotation on the spot and forward flight while turning. The compensation should effectively minimize the influence of wind on the aircraft in a direct as possible way.

1.3. State of the art / Related work

1.4. Airship squid

This work is performed on the airship squid, which is further specified in this section. The airship was designed and built by roboloon and consists of a elastic hull, which is usually filled with Helium. For maneuvering the airships has one stern propeller at the back, four fins and four impellers integrated in the fins totalling 9 control surfaces making the airship an overactuated system.

A full table showing the specifications of the airship can be found in A.1

1.4.1. Hardware

The airship is equipped with an IMU, GPS and a 1D pitot tube measuring the airspeed in the body-fixed x axis. This pitot tube yields the restriction of only measuring airspeeds above 1.5m/s, below this threshold no measurement is available.

1.4.2. Software

The airship is controlled by a PID controller ... The PID is tuned to allow for robust behavior in all flight envelopes.

Allocator

1.5. Simulation Environment / Systematic

All algorithms are first implemented in simulation. This simulation poses a model of the airship as derived in Chapter and an implementation of the PID controller. Additionally uncertainties for the forces are considered. After successful validation in the simulation a real world flight test is carried out.

2. Fundamentals

2.1. Coordinate Systems

For this work two coordinate systems are considered. Firstly the body-fixed coordinate system, which axes align with the body axes according to the North-East-Down (NED) convention. The origin of the body-fixed CS coincides with the Center of Gravity of the Airship.

Secondly the earth coordinate system. For simplicity a flat earth model is assumed. In this coordinate system . The origin refers to an arbitrary point on the earths' surface and is chosen to be at the initial position of the airship. The Earth CS is inertial, which means

why flat earth?

what?

add graphics

add transformation matrix

2.2. Airship Model

The airship is modeled using a six degree-of-freedom (6DOF) Newton Euler approach. Therefore we assume the airship to be a rigid body with constant mass and model the translational and rotational dynamics accordingly. We express the equations of motion in body-coordinates, allowing for a constant inertia matrix. Unless stated otherwise, all quantities are expressed in the body-fixed coordinate frame, and all forces and moments are assumed to act at the center of gravity (COG). The corresponding frame identifier $(\cdot)$ and point-of-application identifier $(\cdot)^{\text{COG}}$ are omitted throughout the remainder of this section for improved readability.

The three-dimensional **translational dynamics** in body-coordinates are given by the linear momentum theorem as

$$\begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & m \end{bmatrix} \mathbf{a}_B = \mathbf{F}_{\text{ext}}, \quad (2.1)$$

$$m \mathbf{I}_{3 \times 3} \mathbf{a}_B = \mathbf{F}_{\text{ext}}, \quad (2.2)$$

where m denotes the mass of the airship, $\mathbf{I}_{3 \times 3} \in \mathbb{R}^{3 \times 3}$ denotes the identity matrix, $\mathbf{a}_B \in \mathbb{R}^3$ describes the acceleration of the airship in body-fixed coordinates and $\mathbf{F}_{\text{ext}} \in \mathbb{R}^3$ represents the resultant external force acting at the center of gravity (COG), expressed in body-fixed coordinates.

To relate the body-fixed acceleration $\mathbf{a}_B$ to the body-fixed velocity $\mathbf{v}_B \in \mathbb{R}^3$, the rotation of the body-fixed coordinate system relative to the earth-fixed coordinate system must be taken into account. Applying Euler's transport theorem yields

$$\mathbf{a}_B = \left(\frac{d}{dt} \mathbf{v}_B \right) = \dot{\mathbf{v}}_B + (\boldsymbol{\omega}_{EB} \times \mathbf{v}_B) = \dot{\mathbf{v}}_B + \mathbf{a}_{\text{cor}}, \quad (2.3)$$

2. Fundamentals

where $\mathbf{a}_{\text{cor}} \in \mathbb{R}^3$ denotes the Coriolis acceleration as a result of the rotating body-fixed CS and $\boldsymbol{\omega}_{EB} \in \mathbb{R}^3$ describes the rotational velocity of the body-fixed CS against the Earth CS. As the Earth CS is inertial this reduces to the angular velocity of the airship along its three axes, $\boldsymbol{\omega}_B \in \mathbb{R}^3$. Inserting equation (2.3) in (2.2) and defining the Coriolis force as

$$\mathbf{F}_{\text{cor}} = \begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & m \end{bmatrix} \mathbf{a}_{\text{cor}} \quad (2.4)$$

yields the dynamics for the translational motion of the airship in body-fixed coordinates

$$\begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & m \end{bmatrix} \dot{\mathbf{v}}_B = \mathbf{F}_{\text{ext}} - \mathbf{F}_{\text{cor}}. \quad (2.5)$$

The three-dimensional **rotational dynamics** in body-fixed coordinates are given by the angular momentum theorem as

$$\left(\frac{d}{dt} (\mathbf{I}_B \boldsymbol{\omega}_B) \right) = \mathbf{M}_{\text{ext}}, \quad (2.6)$$

where $\mathbf{I}_B \in \mathbb{R}^{3 \times 3}$ denotes the inertia matrix of the airship expressed in body-fixed coordinates and $\mathbf{M}_{\text{ext}} \in \mathbb{R}^3$ represents the resultant external moment acting about the center of gravity (COG), expressed in body-fixed coordinates.

To relate the change of angular momentum to the angular acceleration of the airship, the rotation of the body-fixed coordinate system relative to the earth-fixed coordinate system must be considered. Applying Euler's transport theorem to the angular momentum vector yields

$$\left(\frac{d}{dt} (\mathbf{I}_B \boldsymbol{\omega}_B) \right) = \mathbf{I}_B \dot{\boldsymbol{\omega}}_B + (\boldsymbol{\omega}_B \times \mathbf{I}_B \boldsymbol{\omega}_B), \quad (2.7)$$

where the second term represents the gyroscopic acceleration caused by the rotation of the body-fixed coordinate system. Since the inertia matrix is constant when expressed in body-fixed coordinates, the time derivative only applies to the angular velocity.

Defining the gyroscopic moment as

$$\mathbf{M}_{\text{gyro}} = (\boldsymbol{\omega}_B \times \mathbf{I}_B \boldsymbol{\omega}_B) \quad (2.8)$$

and rearranging equation (2.6) yields the rotational dynamics of the airship in body-fixed coordinates

$$\mathbf{I}_B \dot{\boldsymbol{\omega}}_B = \mathbf{M}_{\text{ext}} - \mathbf{M}_{\text{gyro}}. \quad (2.9)$$

We can combine the translational dynamics (2.5) and the rotational dynamics (2.9) into one equation obtaining the Newton Euler Equations for the airship.

$$\mathbf{M} \dot{\boldsymbol{\nu}} = \mathcal{F}_{\text{ext}} - \mathcal{F}_{\text{cor}} \quad (2.10)$$

with the generalized velocity vector $\boldsymbol{\nu}$ and mass matrix $\mathbf{M}$ defined as

$$\boldsymbol{\nu} = \begin{pmatrix} \mathbf{v}_B \\ \boldsymbol{\omega}_B \end{pmatrix} \in \mathbb{R}^6, \quad \mathbf{M} = \begin{pmatrix} m\mathbf{I}_{3 \times 3} & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & \mathbf{I}_B \end{pmatrix} \in \mathbb{R}^{6 \times 6}. \quad (2.11)$$

The generalized external and Coriolis force vectors are defined as

$$\mathcal{F}_{\text{ext}} = \begin{pmatrix} \mathbf{F}_{\text{ext}} \\ \mathbf{M}_{\text{ext}} \end{pmatrix} \in \mathbb{R}^6, \quad \mathcal{F}_{\text{cor}} = \begin{pmatrix} \mathbf{F}_{\text{cor}} \\ \mathbf{M}_{\text{cor}} \end{pmatrix} \in \mathbb{R}^6. \quad (2.12)$$

The generalized external forces acting on the airship can be further classified based on the available knowledge of the airship system. Since the airship is modeled as a rigid body, the internal deformations and distributed effects of the structure are neglected, which represents a significant simplification of the real system. Consequently, the accuracy of the airship model depends strongly on the precise representation of the external forces and moments acting on the vehicle. To capture the relevant physical behavior of the airship, these contributions must therefore be modeled as accurately as possible.

However, due to the complexity of the real environment and the limitations of the available model knowledge, not all forces and moments acting on the airship can be fully accounted for. The resulting discrepancies between modeled and actual system behavior are considered model uncertainties. These uncertainties are not explicitly modeled within the presented equations of motion but are addressed in the subsequent estimation and control design.

The external forces and moments acting on the airship are divided into four main categories: gravitational, buoyancy, aerodynamic, and propulsion.

$$\mathcal{F}_{\text{ext}} = \mathcal{F}_{\text{g}} + \mathcal{F}_{\text{b}} + \mathcal{F}_{\text{aero}} + \mathcal{F}_{\text{prop}}, \quad (2.13)$$

resulting in the final EOM

$$\mathbf{M}\dot{\boldsymbol{\nu}} = \mathcal{F}_{\text{g}} + \mathcal{F}_{\text{b}} + \mathcal{F}_{\text{aero}} + \mathcal{F}_{\text{prop}} - \mathcal{F}_{\text{cor}}. \quad (2.14)$$

The generalized external forces are derived in the following.

2.2.1. Gravitation and Buoyancy

The gravitational force computes to

$$\mathbf{F}_{\text{g}} = m\mathbf{g} = m\mathbf{R}_{\text{BE}} \mathbf{e}_g \quad \text{with} \quad \mathbf{e}_g = \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}, \quad (2.15)$$

where $\mathbf{e}_g$ represents the constant gravity vector in the inertial earth reference frame following the NED convention and $\mathbf{R}_{\text{BE}}$ denotes the transformation matrix from earth CS to body CS.

As the gravitational force acts at the center of gravity, no gravitational moment is induced, thus

$$\mathcal{F}_{\text{g}} = \begin{pmatrix} \mathbf{F}_{\text{g}} \\ \mathbf{0}_{3 \times 1} \end{pmatrix}. \quad (2.16)$$

2. Fundamentals

The buoyancy depends on the density of the surrounding air ρ_{air} and the lifting gas contained within the hull ρ_{gas} as well as the volume of the airship V . Assuming a homogeneous distribution, the resulting buoyancy force acts at the center of volume (COV) and is given by

$$\mathbf{F}_b^{\text{COV}} = -V\mathbf{g}(\rho_{\text{air}} - \rho_{\text{gas}}). \quad (2.17)$$

In case of the airship *Squid*, the volume is chosen such that $\mathbf{F}_g + \mathbf{F}_b > 0$, resulting in a downward-directed net force that prevents the airship from ascending without external input. To express the buoyancy contribution with respect to the center of gravity, the moment induced by the offset between the COV and COG is considered:

$$\mathbf{M}_b = \mathbf{r}_{\text{COV}} \times \mathbf{F}_b^{\text{COV}}, \quad (2.18)$$

where $\mathbf{r}_{\text{COV}} \in \mathbb{R}^3$ denotes the vector from the COG to COV. Due to the design of the airship, $\mathbf{r}_{\text{COV}}$ only contains a component along the z_B -direction. Consequently, the generalized buoyant force expressed at the COG is given by

$$\mathcal{F}_b = \begin{pmatrix} \mathbf{F}_b \\ \mathbf{M}_b \end{pmatrix} \quad \text{with} \quad \mathbf{F}_b = \mathbf{F}_b^{\text{COV}}. \quad (2.19)$$

2.2.2. Aerodynamic

As described in [Li2007AirshipDynamics], the aerodynamic forces and moments can be classified into several contributions, including drag, added mass, viscous effects, and fin forces. Since these contributions account for the majority of the aerodynamic effects, they are considered in the present model.

$$\mathcal{F}_{\text{aero}} = \mathcal{F}_{\text{drag}} + \mathcal{F}_{\text{am}} + \mathcal{F}_{\text{visc}} + \mathcal{F}_{\text{fins}} \quad (2.20)$$

A detailed derivation of the individual contributions is omitted here; instead, the resulting expressions are summarized for completeness. An extensive derivation of the aerodynamic forces and moments can be found in [Li2007AirshipDynamics]. The aerodynamic forces act at the center of volume. Therefore, the offset between the COV and the COG results in an induced moment, which is considered in the aerodynamic moment contribution.

Drag

Added Mass

Viscous Effects

Fins

2.2.3. Propulsion

To be able to compute the Propulsion Components we need to incorporate a Motor Model.

2.3. Wind Model

To make use of these dynamics an accurate wind model is desired. Although the Van Karman Model describes the turbulence spectrum more accurately in most cases this work implements the Dryden Turbulence Model due to its easier implementation. As it results in a linear model. Van Karman is not realizable.

2.3.1. Dryden Turbulence Model

The Dryden wind model can be understood as a random walk with memory. Inputs of the Dryden wind model is white noise. State transition functions

Parameters

The Dryden Model is specified by the following parameters, W_{20} , wind direction and L and sigmas. In the 1970s the US Army developed methods to specify parameters for the Wind model. Therefore we can directly relate the L and sigmas to the current flight state according to following equations:

as W_{20} and wind direction are fully determined by the wind environment these cannot be easily related to the current flight status.

Add comparison plot Random walk and Dryden with appropriate parameter values

2.4. Kalman Filter

Kalman Filters came up in the 19xxs as a method to ... Before that state estimation methods like observers were existent. The first one was the widely popular Luenberger Observer... Kalman Filters offer the advantage to ... they also work especially well in combination with white noise inputs.

Bayesian Filtering

There exist several methods to extend the KF framework to nonlinear systems. 2 of these are explained further in the following section.

2.4.1. Extended Kalman Filter

2.4.2. Unscented Kalman Filter

2.5. Disturbance Feedforward

Disturbance Feedforward describes a method to compensate the influence of a known disturbance on the system. Linear Case Nonlinear Case

3. model-based wind estimation and compensation

There exist several estimation methods in control theory. Beginning from simple linear models like the Luenberger Observer etc... The idea is to combine the knowledge about the nonlinear airships dynamics, which have been developed in 2.2 and the wind model into use a state estimation method. Therefore the Kalman Filter proves to be the perfect solution optimally combining dynamics and measurements.

why KF

3.1. Model

First a complete model is required to calculate the state transition function from it. Therefore we augment the six state airship model including any states that have known dynamics and are measured (position + attitude) and the wind states.

Introduce state vector (12 states) and show that all gen forces can be expressed depending on those states.

State space realization of the wind model. The parameters of the dryden wind model depend on the flight states, so we can easily express them depending on the state. Only the computation of W_{20} and wind direction is not state dependent, this is evaluated in

ref

3.1.1. Discretization

exact discretization of the wind dynamics, approx of model dynamics

3.2. Kalman Filter

To get an Kalman Filter running we need a system in the following form

- discrete state transition function
- discrete measurement function
- Process noise
- Measurement noise

write down
eq, with f ,
 h , Q , R

We consider the case of additive process noise, which becomes clear later as we derive the equations.

3.2.1. discrete state transition function

3.2.2. discrete measurement function

The measurement function directly follows from the sensor equipment on board of the airship

3.2.3. Process Noise and Measurement Noise

we have $Q = [Q_{model}, Q_{wind}]$ uncertainty through unmodeled forces uncertainty coming from wind -> discretization of covariance

measurement noise from sensor specifications Pitot tube only measurements if $u > 2m/s$. To account for this we implement a time varying measurement noise, setting $R_{pitottube} \rightarrow \infty$ if airspeed in x-direction is < 2 . To understand why this works lets look at the computation of the kalman gain. $R \rightarrow \infty$ simply means that we fully trust our dynamics and basically do not use the measurement.

RUKU vs Euler

3.3. Implementation of UKF

The UKF obeys 3 Parameters, which mainly influence the spread of the sigma points. Parameters are set to $\beta = 2$, $\kappa = 0$. The optimal value for α is determined by an optimization run.

3.4. Implementation of EKF

3.4.1. Analytical vs Numerical Jacobian

3.4.2. Robustify EKF

The EKF is not as robust as the UKF due to several reasons. Robustness is a important desired property of the wind estimation algorithm for the specific use case in this thesis, which is to use it for an wind compensation algorithm. Thus to further improve the robustness an Joseph Form computation of the updated covariance is implemented.

3.5. Compensation

While wind is modeled as an additional system state in the Kalman Filter implementation, it can also be considered as an external disturbance acting on the airship. Therefore, the wind estimation provided by the Kalman Filter can be utilized to compensate for the effect of wind disturbances. Two compensation approaches are implemented and evaluated: a nonlinear Disturbance Feedforward (DFF) controller and a Nonlinear Dynamic Inversion (NDI) controller. Both approaches can be interpreted as an augmentation to the existing PID controller, adding an additional compensation term

$$\mathbf{u} = \mathbf{u}_{PID} + \mathbf{u}_{comp}, \quad (3.1)$$

where $\mathbf{u}_{PID}$ is mainly responsible for setpoint tracking and stabilization, while $\mathbf{u}_{comp}$ uses model knowledge of the airship to counteract the wind-induced disturbances in case of DFF, or compensates for all acting forces and moments in case of the NDI.

To minimize the delay between wind disturbance estimation and compensation, the estimated disturbance is compensated directly at the force and moment level. Based on the estimated wind and the current airship state measurement, the required compensation force and moment vector

$$\mathcal{F}_{comp} = \begin{pmatrix} \mathbf{F}_{comp} \\ \mathbf{M}_{comp} \end{pmatrix} \in \mathbb{R}^6$$

is calculated. The computation of $\mathcal{F}_{\text{comp}}$ depends on the compensation approach and is derived in the sections 3.5.1 and 3.5.2.

The total desired force and moment vector is obtained by combining the force and moment vector generated by the PID controller, $\mathcal{F}_{\text{PID}}$, with the compensation vector, $\mathcal{F}_{\text{comp}}$:

$$\mathcal{F} = \mathcal{F}_{\text{PID}} + \mathcal{F}_{\text{comp}}. \quad (3.2)$$

This force and moment vector is subsequently converted into individual motor commands by the control allocation algorithm. Performing the compensation directly at the force and moment level enables computationally efficient execution, as the mapping from forces and moments to motor commands u is performed only once per time step. At the same time, the underlying concept introduced in Eq.(3.1) is preserved, allowing disturbances to be compensated with a delay of only one control cycle.

The resulting control structure, including the compensation strategy, is illustrated in Figure 3.1.

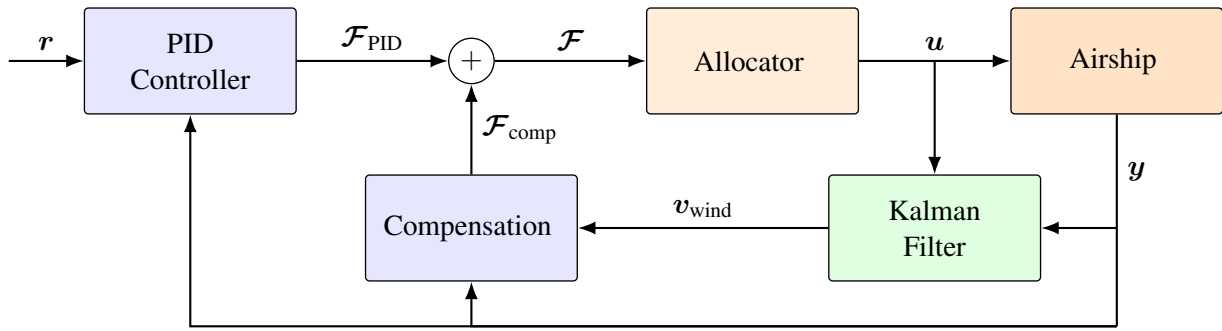


Figure 3.1.: Control architecture consisting of a PID controller, a Kalman filter for wind estimation, and a feedforward disturbance compensation module.

The original PID controller, without compensation, was designed with a strong emphasis on robustness against modeling uncertainties and wind disturbances. Since the compensation approaches explicitly account for a significant portion of these disturbances, the controller can theoretically be retuned to place greater emphasis on tracking performance. This results in a more aggressively tuned PID controller. In section [the effects are evaluated](#).

ref

In the following sections, the two compensation approaches are described in detail, and the derivation of the compensation vector $\mathcal{F}_{\text{comp}}$ is presented.

3.5.1. Disturbance Feedforward

The Disturbance Feedforward compensates for the wind induced forces and moments only. This means $\mathcal{F}_{\text{comp}}$ compensates for the additional forces and Moments created by the prevailing wind. Looking at the dynamics one can see that wind affects our airship only via the aerodynamical forces, as they depend on the airspeed V_a which is

$$V_a = V_b - V_w \quad (3.3)$$

Fprop describes the propulsion force that is achieved by a command u . Faero is the aerodynamical force. In general Faero and Fprop describe nonlinear functions mapping u and V_a to Forces. Fprop can be directly expressed by $\mathcal{F}$

add dynamics, what is Faero and Fprop?

3. model-based wind estimation and compensation

we basically have everything expressed in forces.

The Disturbance Feedforward task is to find u_{comp} s.t.

$$F_{aero}(V_A) + F_{prop}(u + u_c) = F_{aero}(V_B) + F_{prop}(u) \quad (3.4)$$

To further understand what this means lets look how this unfolds assuming a linear system.

$$F(V_A) + F_{prop}(u + u_c) = F(V_B) - F(V_w) + F_{prop}(u) + F_{prop}(u_c) \quad (3.5)$$

$$= F(V_B) + F_{prop}(u), \quad (3.6)$$

$$\Rightarrow F_{prop}(u_c) = F(V_w) \quad (3.7)$$

in this case $F(v_w)$ simply describes the additional forces induced by wind. For nonlinear systems its more complicated.

ref Tuning: The PID controller is retuned on the undisturbed model, as the compensation should eliminate those, allowing for a faster PID control. In the performance of the retuned PID controller is examined and compared to the control structure with only the PID. It shows that the retuned PID is far too aggressive and not able to achieve good performance for all flight envelopes, although performing better on some. The reason for this is that although theoretically the Compensation should compensate for the wind, it is not able to fully cancel it due to uncertainties in the wind estimation and state measurements. Hence a robust controller is desired anyways. A combination of the robust PID with the compensation is beneficial.

3.5.2. Nonlinear Dynamic Inversion

As second approach a Nonlinear Dynamic Inversion (NDI) controller is implemented. The idea behind the NDI is to fully compensate the Forces acting on the Airship, basically inverting the dynamics.

$$todo \quad (3.8)$$

For NDi the model was tuned on the unforced model (so total forces were set to zero). This allows for an pretty fast PID. But again under real conditions this does not provide good results. As there is too much uncertainty in it.

Given perfect measurements and wind estimate the NDI achieves perfect compensation of all acting forces.

4. data-driven wind estimation and compensation

5. Evaluation and results

First the Performance of the Wind estimation algorithm is assessed in simulation. Later the combination of wind estimation and compensation is compared to the flight envelope without compensation. To evaluate the performance of the Estimation and compensation following flight scenarios are considered

- Forward Flight
 - constant velocity
 - acceleration
- Sideway Flight
 - constant velocity
 - acceleration
- Pure rotation around the yaw axis
- turn while flying forward (Vorgabe von Vorwärtsgeschwindigkeit und Drehgeschwindigkeit)

5.1. Wind estimation

Following the wind estimation performance is analyzed. First of all we need to find a $W20$ parameter. We choose $W20 = 5$, which is justified in the section .

ref

Result: Good wind estimation....

5.1.1. Sensitivity to wind parameters

The wind parameters $W20$ and wind direction cannot be measured or computed directly from the states and thus need to be estimated. In the following the sensitivity to the correct parameter is assessed. Therefore multiple tests are carried out.

First a constant parameter value for $W20$ was specified, creating the actual wind turbulences in the simulation. The wind estimation performance was assessed for varying estimated $W20$, reaching from $W20 = 0$ up to $W20 = 30$ km/h. For each estimated $W20$ parameter value 27 wind scenarios are run. In each wind scenario the constant wind, as well as the random seed creating the turbulences, were varied resulting in unique wind conditions for each test. For each test the RMSE between the estimated wind and actual wind are computed individually and the average RMSE along all three axis is computed. Then the results are averaged among all 27 wind scenarios. The results are shown in Table 5.1 and lead to the following conclusions.

a) Even for constant wind $W20_{act} = 0$, an estimate of $W20_{est} = 0$ leads to bad results. This can be explained when looking at the computation of the Wind Process noise, for $W20_{est} = 0$, this results in $q_{wind} = 0$, thus not allowing the wind estimate to change at all. So the initial guess will be kept forever? deviation, due to the nonperfect modelling and uncertainties in place. (For perfect modelling this deviation

proof

really? or
just for
constant
wind = 0

5. Evaluation and results

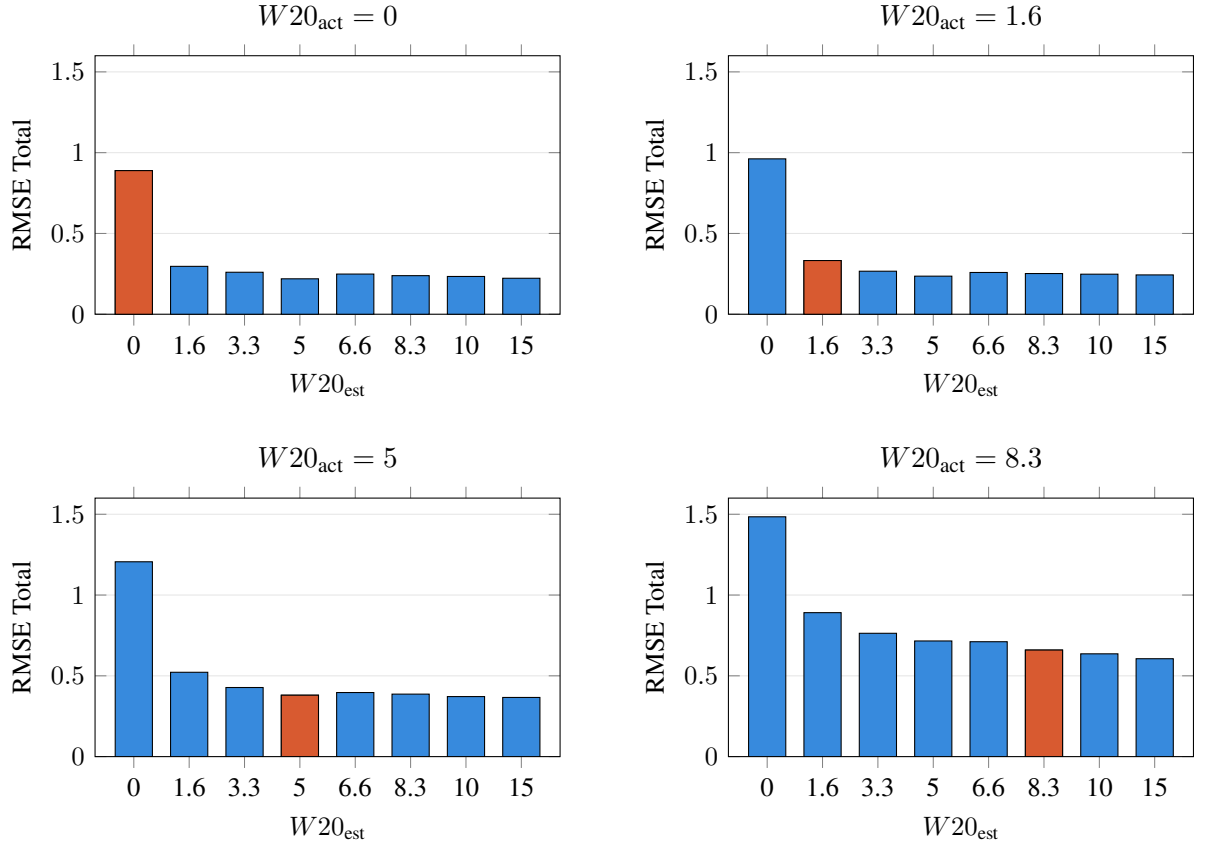


Figure 5.1.: Comparison of Total RMSE Across All Estimated $W20$ Wind Speeds. The orange bar highlights the true configuration scenario where $W20_{est} = W20_{act}$.

would be zero) b) matching $W20$ do not always imply best performance. Reason herefore are errors in modelling and bigger $W20$ more flexibility to adapt. So why not choosing $W20 = 15$ for all? Now look at plots showing differences and oscillations.

The results indicate that a despite the real $W20$ parameter an estimated $W20 = 10$ yields good results for all real wind conditions created by varying $W20$. This can be explained as follows: a bigger $W20$ estimated increases the Process noise, thus giving the wind states more freedom to react to fast changing wind gusts. This comes at a cost which is a more noisy estimation of the wind. $W20$ estimated = 10 seems to offer a good tradeoff although bigger values may lead to an even lower RMSE. And bigger $W20$ leads to bigger q_{wind} , this may have bad side-effects.

5.1.2. Effect of Airspeed Sensor

The Airship is equipped with an 1-D airspeed sensor. Its interesting to know how the wind estimation changes if no airspeed sensor is in place, to quantify the importance of the AS.

5.2. Wind compensation

The wind compensation is heavily dependant on the quality of the wind estimation.

Setting:

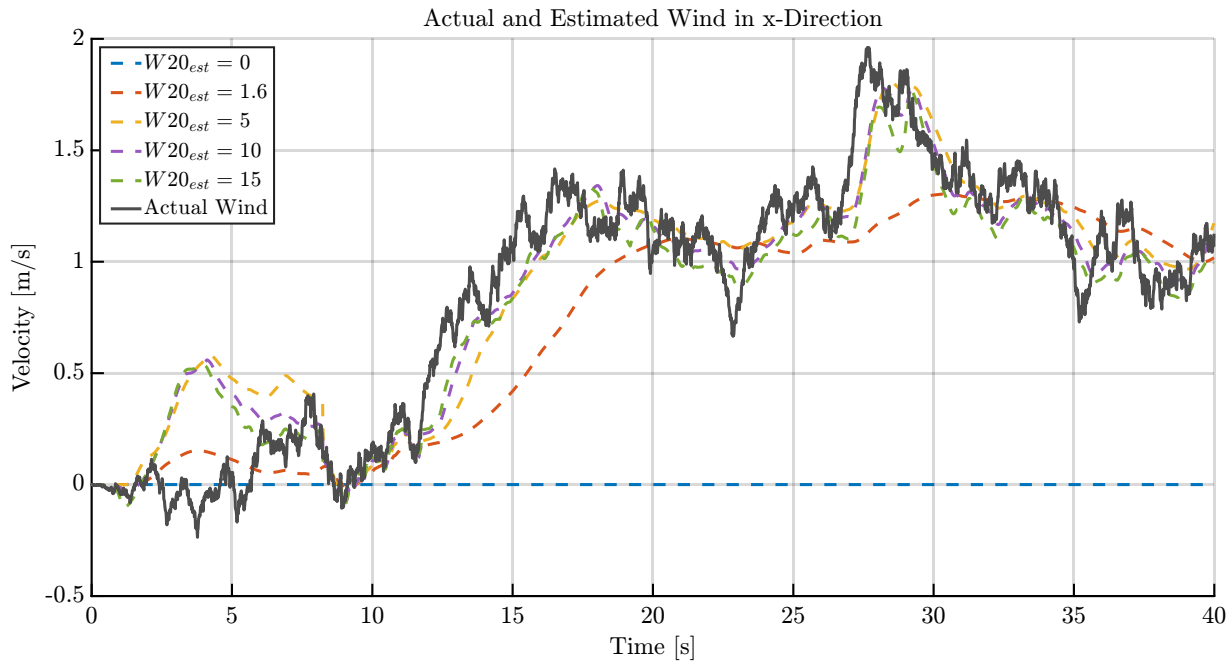


Figure 5.2.: Comparison of wind estimation performance for different estimated $W20$, $W20$ actual is 5.

Results

While theoretically the compensation should allow for a more aggressive PID as the disturbance should compensate the wind disturbance this does not hold up in the results presented. Reason therefore is errors in wind estimation, convergence rates of the KF, maybe even delay of one time step, general uncertainty in the model (uncertain forces etc)

5.2.1. Problems with Observability

Due to the lack of sensory equipment to measure airspeed the wind estimation depends heavily on the effect of the aerodynamic forces on the airship to estimate the wind velocities. Because only the aerodynamic Forces are directly dependent on the airspeed and thus on the wind velocity. The Compensation uses the current wind estimate to compute the estimation and thus requires a good estimation of wind. Exactly there lies the problem, if the compensation fully compensates for the effect of wind on the airship, the wind estimation quality automatically decreases, leading to a poor estimation and so on. To break up this cycle we need to guarantee that the information about the effect of the wind is not fully lost. To achieve this a gain, manipulating the output of the compensation is introduced. This gain in range 0 to 1 determines how aggressively the wind compensation works and balances the wind compensation with the wind estimation.

5.3. Flight Test

6. Conclusion and Outlook

Outlook

The model-based approach proves to be a powerful tool for wind estimation given a good model of the system is available. The Dryden wind model does not only provide a mathematical description for the wind speeds acting at the center of gravity but also yields a description for the effect of the wind on the angular rates, which are caused by differences in wind magnitudes within the prevailing wind field.

Including the angular rates induced by the differences in wind speeds within the wind field can improve the performance as the airship is especially vulnerable to wind fields considering the airship's large side surfaces. Angular rates of the wind are described by transition functions of order 2 (for the u-direction) and order 3 (for v, w direction) resulting in a increase of one order for every direction compared to wind speeds. It remains to investigate whether the gain in performance can justify the increased complexity.

The Wind estimation proved to be particularly reliable. This information can be used for advanced control and trajectory planning algorithms like MPC to increase the performance compared to the current PID setup.

Bibliography

- [1] Yuwen Li and Meyer Nahon. “Modeling and Simulation of Airship Dynamics”. In: *Journal of Guidance, Control, and Dynamics* 30.6 (2007), pp. 1691–1700. doi: 10.2514/1.29061. URL: <https://doi.org/10.2514/1.29061>.

A. First Appendix

Write your appendix content here.

A.1. Parameters of the airship squid

Table A.1.: Example Airship Parameters and Specifications

Parameter	Symbol	Value	Unit
Overall length	L	75	m
Maximum diameter	D	18	m
Envelope volume	V	10 000	m ³
Total mass	m	8 500	kg
Payload mass	m_p	1 500	kg
Helium mass	m_{He}	1 700	kg
Cruise speed	v_c	90	km/h
Maximum speed	$v_{\max}$	120	km/h
Maximum altitude	$h_{\max}$	3 000	m
Endurance	t_e	24	h
Propulsion power	P	2×150	kW
Number of engines	N_e	2	–
Buoyant force	F_b	98 000	N
Drag coefficient	C_D	0.045	–
Reference area	A	255	m ²
Moment of inertia (roll)	I_{xx}	1.2×10^5	kg m ²
Moment of inertia (pitch)	I_{yy}	8.5×10^5	kg m ²
Moment of inertia (yaw)	I_{zz}	8.9×10^5	kg m ²
Center of buoyancy offset	z_B	0.8	m

B. Second Appendix

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.